CGIAR

CGIAR SCALING FOR IMPACT

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Leveraging Public-Private Rice Partnership Models for Food and Nutrition Security in Côte d'Ivoire and Nigeria

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.2 · Entrepreneurship through Mechanization & Postharvest Services

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AfricaRice

Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

  Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

  Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

  Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

  Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

  Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

  Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

West Africa's rice demand is rapidly increasing, yet countries like Côte d'Ivoire and Nigeria remain heavily import-dependent due to systemic bottlenecks. These include low adoption of climate-resilient varieties, weak seed systems, limited mechanization services, and postharvest losses exceeding 20-30%. Current interventions are often fragmented and fail to reach scale, while climate variability and market volatility compound risks for smallholders. This Scaling Track addresses these challenges by transforming the rice value chain into a resilient, competitive, and inclusive system that contributes to national food sovereignty.

The primary goal is to scale a comprehensive innovation bundle across two key geographies: Côte d'Ivoire (anchored by the AfricaRice HQ and the national REWARD program) and Nigeria (via sub-national public-private partnerships in Niger State and Jigawa State). The solution is not a single technology but a **system innovation bundle** delivered through state-level Public-Private Partnerships (PPPs). It integrates climate-smart seeds, mechanization-as-a-service, post-harvest processing improvements, nutrition-sensitive value addition, and a digital backbone for accountability and advisory services.

The innovations are ready for scale, having been tested through the TAAT, REWARD, and HealthyDiets4Africa programs. AfricaRice varieties have already reached over 2 million farmers, and the GEM parboiling system has increased incomes for women's groups by 20-30%. The Scaling for Impact (S4I) investment will act as a catalyst to bundle these proven solutions, embed digital monitoring, and leverage significant counterpart financing from government and private sector partners to achieve systemic change.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

- **Public-Private Seed Systems:** Stewardship of breeder and foundation seed by public institutes (AfricaRice, CNRA, NASC) linked with commercial certified seed production and distribution by licensed private seed companies.
- **Mechanization-as-a-Service (MaaS):** Youth- and SME-led service hubs providing affordable, timely access to land preparation, planting, harvesting, and milling on a pay-for-service basis.
- **Post-Harvest and Quality Innovations:** Scaling of energy-efficient parboilers (GEM), solar/rice husk dryers, and improved milling to reduce losses, lower GHG emissions, and improve grain quality for premium markets.
- **Nutrition-Sensitive Value Addition:** Production and marketing of low-glycemic index (GI) parboiled rice and fortified rice products, led by women and youth enterprises to target healthier urban and rural diets.
- **Digital Backbone (eProd Platform):** Farmer registration and GPS mapping, real-time traceability of inputs and paddy, SMS-based climate and agronomic advisories, and performance monitoring dashboards.

2.2 Evidence of Impact

- **Yield Increase:** Pilots in Nigeria and Côte d'Ivoire show that mechanization and improved practices increase yields by **25–40%** .
- **Income Growth:** The GEM parboiling system has improved incomes for over **500 women's groups** by **20-30%** .
- **Digital Reach:** Digital advisory (RiceAdvice) has reached **80,000 farmers**, boosting yields by **0.5-1.5 t/ha**. The eProd platform has already registered **14,366 farmers** and mapped **12,110 ha** under TAAT.

- **Scaled Production:** In Nigeria, the Niger Foods PPP mobilized **100,000 farmers** and mechanized **50,000 ha in 2024**, securing structured offtake agreements for **1.75 million tons** of paddy annually with Dangote and other large processors.
- **Seed Delivery:** The PPP model successfully delivered large volumes of certified seed in Liberia and Senegal under an output-based incentive scheme, demonstrating clear economic incentives for private sector participation.

2.3 Scaling Targets

Outcome Metric	2026 Target	Long-term Goal (3–5 years, by 2028)
Farmers reached	500,000 farmers registered on eProd	1 million farmers (40% women, 30% youth)
Area under improved practices (ha)	300,000 ha (cumulative)	500,000+ ha (cumulative)
Jobs supported	20,000 new jobs created	50,000 new jobs along the value chain
Postharvest loss reduction	Baseline established; 10% reduction	25% reduction in postharvest losses
Investment leveraged	USD 50 million	USD 200+ million from government/private sector
Structured offtake (Nigeria)	1.0 million tons paddy	1.75 million tons paddy annually

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy

The scaling pathway is built on a **public-private partnership (PPP) model** that aligns the economic incentives of all actors. The core strategy is to move beyond fragmented pilots by embedding the innovation bundle within existing, large-scale state and private investment programs. Instead of creating new structures, the initiative strengthens and links existing platforms. The pathway relies on three pillars:

- **Public Sector Enabling Role:** Governments provide policy support for seed certification, anchor investments in irrigation (e.g., REWARD), and provide stewardship of early-generation seed.
- **Private Sector as Delivery Engine:** Private seed companies, millers (e.g., Dangote, TGI), and mechanization SMEs deliver services and products as viable, profitable businesses.
- **Digital Integration for Accountability:** The eProd platform provides the "glue" that enables traceability from seed to market, monitors performance, and provides climate advisories, ensuring transparency and reducing risk for all partners.

3.2 Theory of Change

If smallholder farmers, women processors, and youth entrepreneurs gain access to a bundled package of quality seeds, mechanization services, post-harvest technologies, and digital tools through functional public-private partnerships, **and if** key market and coordination failures (e.g., seed fragmentation, lack of service markets, poor traceability) are addressed through the eProd digital backbone and structured offtake agreements, **then** immediate outcomes will include increased yields, reduced postharvest losses, and higher incomes, **leading to** intermediate outcomes such as the emergence of viable, private-sector-led service enterprises and expanded local rice production, **ultimately contributing to** a long-term, systemic shift towards a resilient, competitive, and import-substituting rice value chain in Côte d'Ivoire and Nigeria.

Key Activities Funded by S4I

- **Bundling and Technical Assistance:** Co-design of the innovation bundle with partners; training-of-trainers on GAP, mechanization, and digital literacy; technical backstopping for seed production and processing clusters.
- **Digital Integration:** Deployment and customization of the eProd platform for partner use; support for farmer registration and GPS mapping; integration of climate and weather advisory services.
- **Partnership and Governance:** Facilitation of PPP agreements and structured offtake contracts; support for policy dialogue on seed certification and mechanization standards.
- **Learning and Safeguards:** Establishment of a robust MEL system using eProd and MPRO tools; targeted support to ensure 40% women and 30% youth participation.

4. Partners and Roles

Partner Type	Specific Partners and Quantities	Role in Scaling Pathway
Demand Partners	**Private Sector:** Niger Foods Ltd., Dangote Group, TGI, Lagos Produce Project (Nigeria); Africa-wide Seed and Miller's Network; MARUO Farms (Gambia); local mechanization SMEs and youth groups (est. 4,000+ by 2028). **Farmers:** 1 million smallholders (target).	**Primary drivers of scale.** They invest in production, provide services, and create market demand. They are the end-users and financiers of the bundled innovations, ensuring long-term sustainability through revenue generation (e.g., seed sales, service fees, paddy offtake).
Scaling Partners	**Government and National Institutions:** Ministries of Agriculture (Côte d'Ivoire, Nigeria); Niger and Jigawa State Governments; CNRA, ADERIZ, NASC, NARI. **CGIAR and Development Programs:** AfricaRice (lead), CIMMYT (mechanization); AfDB (REWARD, TAAT), IsDB (RVCDP), World Bank (VIVA-Benoue).	**Enablers and anchors.** They provide policy support, co-financing, foundational infrastructure (irrigation, roads), and technical stewardship. They ensure the initiative is aligned with national strategies and embedded in large-scale investment programs, providing the platform for scale.

Funding Model: S4I funding provides **catalytic support** for bundling, learning, digital integration, and safeguards, while **core production infrastructure, seed multiplication, and service costs are financed by state budgets, bilateral programs (AfDB, World Bank), and private sector counterpart financing.**

5. Enabling Environment and Risks

5.1 Opportunities

- **Strong Government Demand:** National strategies (ECOWAS Rice Offensive, National Rice Development Strategies) and high-level political will (e.g., Niger State Government's commitment to 400,000 ha) create a favorable policy environment.
- **Significant Committed Co-Financing:** The initiative leverages substantial, existing investments from AfDB (REWARD), IsDB (RVCDP), World Bank (VIVA-Benoue), and state governments (approx. USD 1.9B in Nigeria), reducing reliance on short-term donor funding.
- **Private Sector Appetite:** Major corporations like Dangote are actively seeking structured offtake agreements to secure local paddy supply, creating a powerful market pull for the innovation bundle.
- **Proven Digital Infrastructure:** The eProd platform is already operational and tested, providing a ready-made tool for traceability and accountability that can be rapidly deployed.

5.2 Challenges and Mitigation

- **Political Economy and Import Lobby:** Vested interests from rice importers may resist efforts to boost local production. **Mitigation:** Local partners (e.g., Niger Foods, political champions) are positioned to manage this; the focus on improving local rice *quality and competitiveness* addresses the root cause of consumer preference for imports.
- **Insecurity and Instability:** Conflict in parts of Northern Nigeria poses an operational risk. **Mitigation:** Activities will be focused in lower-risk areas identified by state partners; implementation will rely on strong local networks to minimize direct exposure.
- **Mechanization Sustainability:** Spare parts and repair services are often unavailable, leading to equipment downtime. **Mitigation:** A new agreement with a Thai consortium will establish local service and repair hubs, promoting the use of more suitable, smaller-scale equipment.
- **Digital Adoption:** Limited connectivity and digital literacy may hinder platform use. **Mitigation:** A hybrid model will combine the app with SMS-based services and the use of local field agents for data capture and farmer support.
- **Financial Access:** SMEs and farmers lack credit. **Mitigation:** Linkage with blended finance and guarantee schemes within the REWARD and Niger Foods programs; the pay-for-service and offtake models themselves reduce financial risk by ensuring revenue streams.

6. Learning and Adaptation

6.1 Monitoring Focus

- **Process Outcomes:** Functionality of PPP models; number of active service providers; volume of certified seed flowing through private channels; farmer registration and GPS mapping rates.
- **Behavior Change:** Adoption rates of bundled innovations (disaggregated by gender/youth); shifts in planting and harvesting timeliness; use of digital advisories.
- **System Outcomes:** Reduction in postharvest losses; paddy prices and quality improvements; volume of local rice entering structured markets; partner co-investment leveraged.
- **Inclusion:** Verification of 40% women and 30% youth participation in all activities and benefits, tracked via eProd's gender-disaggregated data.

6.2 Adaptive Management

Learning will be real-time and drive adaptive management. The eProd dashboard and the AfDB MPRO tool will provide live data for quarterly reviews with the S4I team. These reviews will assess progress against targets and identify bottlenecks (e.g., low service uptake in a specific area, equipment breakdowns). Findings will trigger rapid, on-the-ground problem-solving with partners. Formal mid-term and annual reviews will be used for more strategic course corrections and to document evidence for replication and future investment.

7. Next Steps and Timeline (2026)

Q1	Finalize partnership agreements with Niger Foods, Jigawa State, and ADERIZ. Initiate large-scale farmer registration and GPS mapping via eProd. Begin large-scale demonstrations of high-yielding, climate-smart varieties at all hub sites. Conduct initial training-of-trainers for mechanization SMEs.
Q2	Launch first cohort of youth-led mechanization service hubs. Distribute breeder and foundation seeds to licensed private seed companies. Roll out SMS-based climate and GAP advisories via eProd. Conduct gender and youth inclusion baseline survey.
Q3	Expand eProd registration to 500,000 farmers. Establish first post-harvest processing clusters (GEM parboilers, dryers) in Côte d'Ivoire. Operationalize structured offtake agreements in Nigeria for the harvest season. Hold mid-year review with all partners.
Q4	Document and disseminate initial evidence on cost-effectiveness and responsible scaling. Consolidate lessons learned for 2027 scaling plan. Validate 2026 achievement data (farmers reached, area covered, jobs created).

Communications and Knowledge: Key products will include quarterly briefs with dashboard snapshots, success stories focused on youth and women entrepreneurs, and a mid-year learning report. Channels will include the S4I community of practice, partner workshops, and targeted policy briefs for national ministries.

8. Contact

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